GOVERNMENT OF KHYBER PAKHTUNKHWA

KHYBER PAKHTUNKHWA TEXTBOOK BOARD PESHAWAR

OFFICIAL PRESCRIBED E-CURRICULUM TEXTBOOK

BIOLOGY CLASS 10

CLASS / GRADE: 10th (Matric)

Subject: Biology | Language: English

AUTHENTIC CURRICULUM SYLLABUS & CHAPTER BLUEPRINTS

· Total Complete Curriculum Chapters: 9 Units

· Designed as per National Curriculum & KPK BISEP Standards

· Contains complete theory, conceptual exercises, MCQs & solutions

· Fully synchronized with ExamForge AI Examination Paper Generator

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UNIT 1: GASEOUS EXCHANGE: IN PLANTS AND RESPIRATORY SYSTEM

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Gaseous Exchange: In Plants and Respiratory System in Humans.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
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UNIT 2: HOMEOSTASIS: OSMOREGULATION, THERMOREGULATION, EXC

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Homeostasis: Osmoregulation, Thermoregulation, Excretory System

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

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[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 3: COORDINATION AND CONTROL: NERVOUS SYSTEM, REFLEX A

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Coordination and Control: Nervous System, Reflex Action, Endocrin

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 4: SUPPORT AND MOVEMENT: HUMAN SKELETON, JOINTS, MUSC

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Support and Movement: Human Skeleton, Joints, Muscles & Movem

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 5: REPRODUCTION: ASEXUAL AND SEXUAL REPRODUCTION IN P

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Reproduction: Asexual and Sexual Reproduction in Plants and Anim

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 6: INHERITANCE: MENDEL'S LAWS, CHROMOSOMES AND GENES,

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Inheritance: Mendel's Laws, Chromosomes and Genes, DNA Replic

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 7: MAN AND HIS ENVIRONMENT: ECOSYSTEM, FOOD WEB, BIOG

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Man and His Environment: Ecosystem, Food Web, Biogeochemical

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 8: BIOTECHNOLOGY: FERMENTATION, GENETIC ENGINEERING,

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Biotechnology: Fermentation, Genetic Engineering, Insulin Productio

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 9: PHARMACOLOGY: MEDICINAL DRUGS, ANTIBIOTICS, VACCIN

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Pharmacology: Medicinal Drugs, Antibiotics, Vaccines, Sedatives &

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

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